Data handling & Visualization | Activity-5 | 15-10-2026 | Y. Veerendar Reddy, 192325102, Slot-C.

PART-A

Chart:1: Resting Pulse Cbeats per min); n>60:-

* Normal ? → No
* Shape : Right-skewed
* Evidence: The upper tail bends above the reference line with high outlier around 112bpm.

Chart:2: Plant height (cm); n>60:-

* Normal ? → No
* Shape : left-skewed (negative skew)
* Evidence: The points curve away from the reference line, with lower tail above lined upper tail below it.

Chart:3: Daily study time (min), n=30:-

* Normal ? → No
* shape : Right-skewed
* Evidence:- The upper-tail points bend above the reference line, with a large high outlier around 172 min.

chart-1:- Lab experiment duration (min), n>80:-

* Normal ? → Approximately yes.
* shape :→ Approximately symmetric, with a mild right tail.
* Evidence :→ most points follow the reference line closely, although the largest values bend above it & one high value is near 120 min.

Part-B :-

B1) A. chart 2 departs from normality due to left-skewness. Try a suitable transformation, such as squaring the values if then create the Q-Q- plot again.

B2) A R-code :-

```
ggplot (df, aes (Sample = value)) +
    stat_qq() +
    stat_qq_line ).
```